

Revised manuscript submission to Scientific Reports

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To: Dr Osman Orhan, Editorial Board Member

Dear Dr Orhan,

We present our revised manuscript, *Spatial aggregation alters population-weighted estimates of InSAR observation support*, for reconsideration. The original title was *Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow*.

The revised study quantifies how spatial aggregation changes population-weighted InSAR support under fixed inputs. We position this question alongside published population-allocation choices and established InSAR quality control. The contribution is the matched-input evidence, rather than a new covariance identity. Complete-interior comparisons isolate the allocation effect from boundary handling; crossed population-model and pair-selection tests establish its input dependence.

A connected LiCSBAS experiment distinguishes sampled input support from final processing quality and evaluates internal reconstruction using excluded edges. The Fresno analyses test transfer to published observation polygons and assess population-model differences against Census and ACS reporting units. Further analyses quantify estimator availability under clustered missingness, the coverage-residual tradeoff and regional limits to transfer. The smaller regional effects, incomplete residential coverage and mixed GNSS agreement limit the scope of the conclusions. We corrected estimator-availability accounting, DWR interval annualization, population denominators and the interpretation of the block-group benchmark.

We have withdrawn the interpretation of the original 3.77-million figure as a verified count of people exposed to subsidence in unobserved locations. Actual-mask bounds now show what threshold-class allocation can be identified conditionally, without recovering missing motion. A reproducible local GNSS feasibility audit found near-colocated stations and no independently established common datum; we do not present it as Bangkok accuracy validation. The Iran source attribution is corrected and its incomplete analysis excluded.

The accompanying files contain the clean and numbered manuscripts, Supplementary Information, six main figures, a point-by-point response with final PDF locations, and the two portable audit packages. The main text presents six figures and two tables; detailed ranking and evidence summaries accompany the complete results in the Supplementary Information. The scientific reproduction package is available at v0.6.2, which preserves the numerical baseline and includes the current submission documents, the local feasibility and product-support audits, and a section-to-output index. The previous Zenodo DOI covers v0.3.0 only; a revised DOI deposit remains pending.

Sincerely,

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